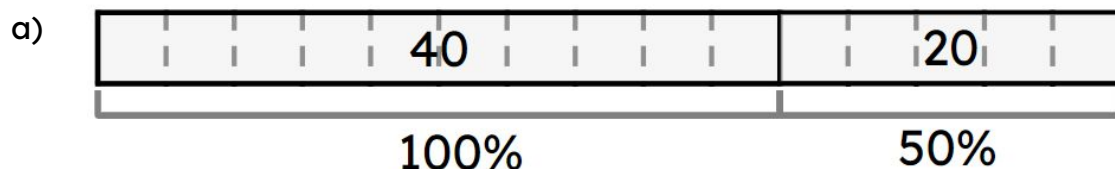




Increase by a percentage

Task A: Increasing an amount by a percentage

1) Write questions and answers for the representation of these bar models:



Possible questions:

Answers:

Increase 40 by 50%

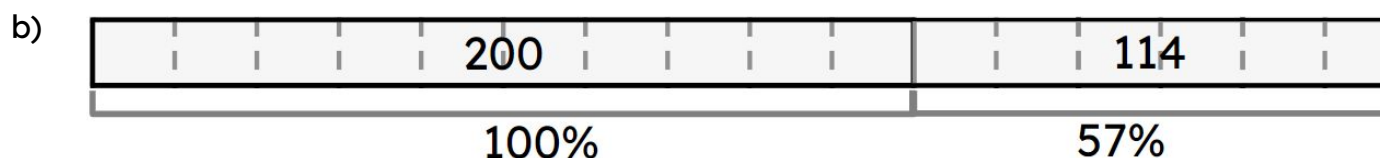
60

Work out 150% of 40

60

What percentage of 40 is 60?

150%



Possible questions:

Answers:

Increase 200 by 57%

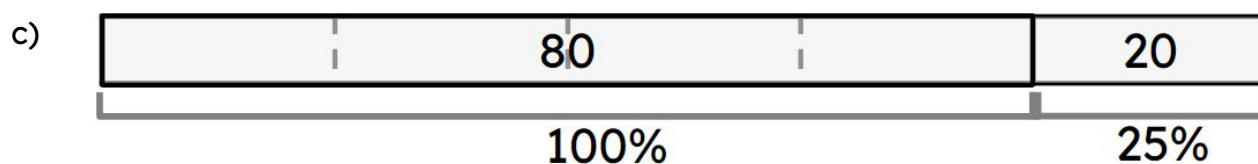
314

Work out 157% of 200

314

What percentage of 200 is 314?

157%



Possible questions:

Answers:

Increase 80 by 25%

100

Work out 125% of 80

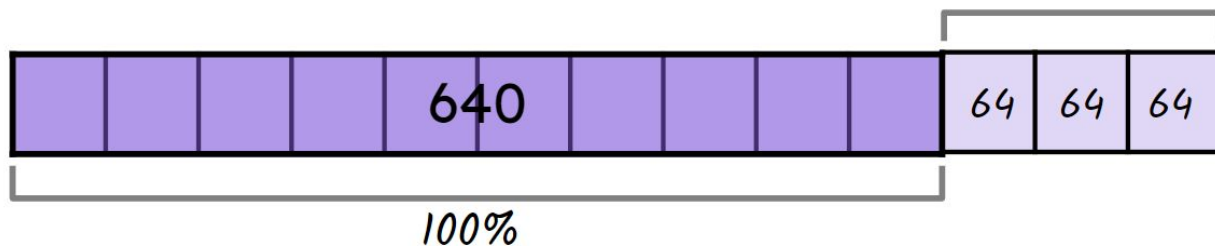
100

What percentage of 80 is 100?

125%

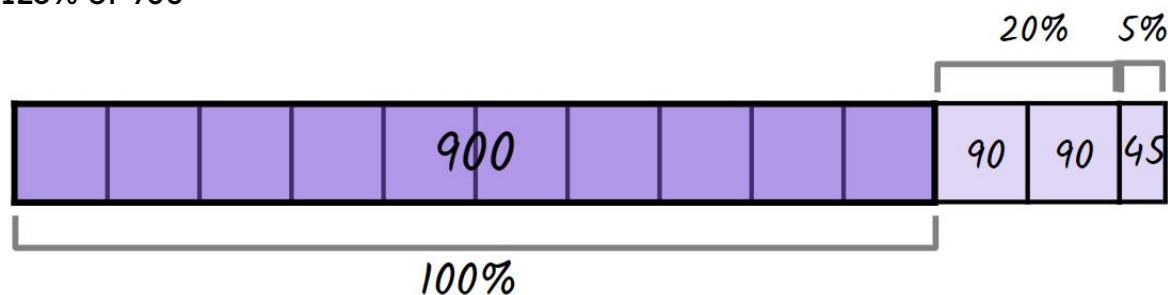
2) Drawing bar models, show how you work out:

a) An increase of 30% to 640



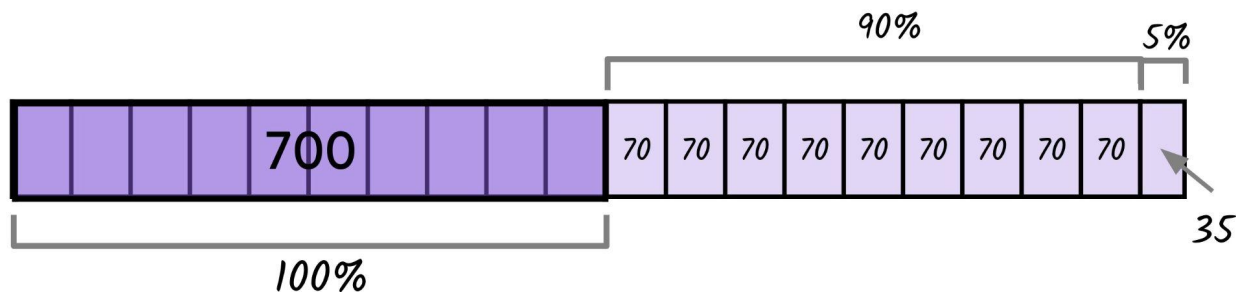
Increasing 640 by 30% is $640 + 64 \times 3 = 832$

b) 125% of 900

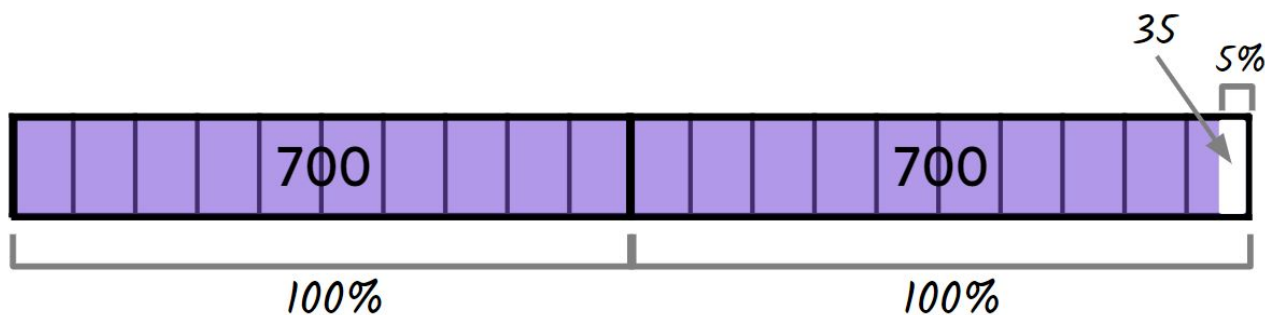


125% of 900 is $900 + 90 \times 2 + 45 = 1125$

c) Increase 700 by 95%



Increase 700 by 95% is $700 + 70 \times 9 + 35 = 1365$



A more efficient way to increase 700 by 95% is to

Add 100% and subtract 5%

$700 + 700 - 35 = 1365$

Task B: Using a multiplier

1) Match the calculation with the question:

Increase 40 by 65%	65×1.65
Work out 125% of 40	40×1.65
Increase 65 by 56%	40×1.25
Work out 156% of 56	65×1.56
Increase 65 by 65%	56×1.56

2) Work out the amount of money when £640 has been increased by 41%

$$640 \times 1.41 = 902.4$$

£640 increased by 41% is £902.40

3) Work out 138% of 700

$$700 \times 1.38 = 966$$

4) Aisha tries to increase £40 by 5% by working out 40×1.5

Explain why she is incorrect and correctly work out the answer.

Aisha worked out an increase of 50% not 5% of 40.

Increasing 40 by 5% is $40 \times 1.05 = 42$ so her answer should be £42

